

Tail Bounds for Sample Median, and Estimation by Median-of-Means

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Abstract

Means of random variables are of ubiquitous interest, but sample means are sensitive in various ways, and are not always the best way to approach mean estimation. Conversely, the median is intuitively rather less sensitive, but appears to estimate the wrong thing. Can we find a productive middle ground?

Basics

I was interested in the concentration properties of the sample median, and the extent to which it has better integrability properties than the ‘seed’ random variable. To make headway on this, one can start by working with tail bounds.

Let $N \in \mathbf{N}$, $M = 2 \cdot N - 1$ and $X_1, \dots, X_M \stackrel{\text{i.i.d.}}{\sim} p$, where $p \in \mathcal{P}(\mathbf{R})$ has cumulative distribution function F . I want to obtain good estimates on the behaviour of $\nu_N := \text{med}(\{X_i : i \in [M]\}) = X_{(N)}$.

To this end, observe that if $X_1, \dots, X_N \geq t$, then $\nu_N \geq t$; it follows that

$$\mathbf{P}(\nu_N \geq t) \geq F(t)^{M-N} \cdot \bar{F}(t)^N.$$

(Actually, the exact value here is given by $\mathbf{P}(\nu_N \geq t) = \binom{M}{N} \cdot F(t)^{M-N} \cdot \bar{F}(t)^N$, which is bigger by a factor $\gtrsim 2^{2N}$.)

If $\bar{F}(t) \in \Omega(t^{-\alpha})$ (i.e. X is polynomially-heavy tailed with tail index $\alpha > 0$), then this implies that $\mathbf{P}(\nu_N \geq t) \in \Omega(t^{-N \cdot \alpha})$. As such, if p has polynomial tails, then $\text{Law}(\nu_N)$ will generally also have polynomial tails for any fixed N . However, the lightness of those tails rapidly improves with N , to the point that the polynomial aspect quickly becomes exponentially negligible.

Orlicz Integrability of X

Follow some slides of Lugosi, let Ψ be a Young function ($\Psi : \mathbf{R}^+ \rightarrow \mathbf{R}^+$, $\Psi(0) = 0$, Ψ is strictly increasing, and convex), and assume the normalisation $\mathbf{E}[\Psi(X)] \leq 1$. Markov’s inequality then yields that

$$t > 0 \quad \rightsquigarrow \quad \mathbf{P}(|X| \geq t) \leq \Psi(t)^{-1}.$$

Inverting this reasoning, for any $\delta \in (0, 1)$, it holds with probability $1 - \delta$ that $|X| \leq \Psi^{-1}(\delta^{-1})$. Moreover, the sample median will satisfy $|\nu_N| \leq \Psi^{-1}(\delta^{-1})$ if at least N of the samples lie within this interval, and so (assuming that $\delta \leq \frac{1}{2} < \frac{N}{M}$), we can estimate that

$$\begin{aligned} \mathbf{P}(|\nu_N| > \Psi^{-1}(\delta^{-1})) &\leq \mathbf{P}(\text{Bin}(M, \delta) \geq N) \\ &\leq \exp\left(-\frac{2 \cdot (N - M \cdot \delta)^2}{M}\right), \end{aligned}$$

where the latter inequality follows from Hoeffding’s inequality. (For δ far from $\frac{1}{2}$, Bernstein’s inequality might yield better bounds.)

As such, if t is sufficiently large (in particular, if $\Psi(t) > 2$), then we have the estimate

$$\begin{aligned}\mathbf{P}(|\nu_N| > t) &\leq \exp\left(-\frac{2 \cdot \left(N - M \cdot \Psi(t)^{-1}\right)^2}{M}\right) \\ &\leq \exp\left(-N \cdot \left(1 - 2 \cdot \Psi(t)^{-1}\right)^2\right).\end{aligned}$$

Worked Example: Finite Variance

For example, if $\Psi(t) = t^2$, then one ends up with $\mathbf{P}(|\nu_N| > t) \leq \exp\left(-N \cdot (1 - 2 \cdot t^{-2})^2\right)$ for $t \geq 2^{\frac{1}{2}}$. For $\delta > \exp(-N)$, one can thus say that with probability $\geq 1 - \delta$, it holds that

$$|\nu_N| \leq \sqrt{\frac{2}{1 - \sqrt{\frac{1}{N} \cdot \log\left(\frac{1}{\delta}\right)}}}.$$

Using that for $0 \leq v \leq \varphi^{-1} := \frac{\sqrt{5}-1}{2}$, $\frac{1}{\sqrt{1-v}} \leq 1 + v$, one can even say that when δ is not too small¹, it holds with probability $\geq 1 - \delta$ that

$$|\nu_N| \leq \sqrt{2} \cdot \left(1 + \sqrt{\frac{1}{N} \cdot \log\left(\frac{1}{\delta}\right)}\right),$$

and this formulation might be more usable. In any case, one can ignore an event of probability $\exp(-\Omega(N))$ and obtain a random variable with ‘qualitatively’ sub-Gaussian tails.

Application to Median-of-Means

This behaviour is put to great use in the Median-of-Means estimator. Given $N = A \cdot B \gg 1$ iid samples $\{X_i : 1 \leq i \leq N\}$, one partitions them into $B > 1$ blocks, each of equal size A , computes the within-block means $\{\mu_j : 1 \leq j \leq B\}$, and then reports the median of the within-block means, which we write as

$$\nu^{\text{MoM}} = \text{median}(\{\mu_j : 1 \leq j \leq B\}).$$

For initial variables $\{X_i\}$ with some integrability properties (e.g. finite variance, higher-order deviations, etc), the block-wise means $\{\mu_j\}$ will typically have even better quantitative regularity as A grows. For example, if $\mathbf{Var}(X_i) = v$, then it is classical that $\mathbf{Var}(\mu_j) = v/A$. As such, taking A larger allows us to work with random variables with both a correct mean and reduced variability. On the other side, taking B larger allows us to compute the median of a larger collection of variables, where we can benefit from the general stabilising quality of the median operator.

In what follows, it helps to conceive of the X_i as being centered. Applying the arguments of the previous section with

$$\nu^\circ = \text{median}\left(\left\{\sqrt{\frac{A}{v}} \cdot \mu_j : 1 \leq j \leq B\right\}\right), \quad \Psi(t) = t^2,$$

one sees that with probability $\geq 1 - \delta$ (provided that $B \geq \log\left(\frac{1}{\delta}\right)$), it holds that

$$|\nu^\circ| \leq \sqrt{2} \cdot \left(1 + \sqrt{\frac{1}{B} \cdot \log\left(\frac{1}{\delta}\right)}\right).$$

and hence that the actual Median-of-Means estimator satisfies

$$|\nu^{\text{MoM}}| \leq \sqrt{\frac{2 \cdot v}{A}} + \sqrt{\frac{2 \cdot v}{N} \cdot \log\left(\frac{1}{\delta}\right)}.$$

¹In particular, $\delta \geq \exp(-\varphi^{-2} \cdot N)$ suffices.

With $B \geq \xi^2 \cdot \log\left(\frac{1}{\delta}\right)$ for some $\xi > 1$, this yields the bound

$$|\nu^{\text{MoM}}| \leq (1 + \xi) \cdot \sqrt{\frac{2 \cdot v}{N} \cdot \log\left(\frac{1}{\delta}\right)}.$$

Note that this bound is valid under only the assumption that X has a finite variance, and yet we obtain uncertainty sets which are analogous to what would usually be obtained for the sample mean under sub-Gaussian assumptions. This apparent paradox can be partially accounted for by noting that i) the estimator implicitly depends on the confidence level δ , ii) this confidence level cannot be taken arbitrarily small.